

Name(s): _____

DATA 101 Assignment 10: Speech text

Work in teams of 2

Instructions. Create a new Jupyter Notebook. Use Markdown cells for all text explanations/headings and Code cells for Python code. Your file should be named `australia.ipynb`. Once finished, export the notebook to HTML via **File → Save and export notebook as → HTML (.html)** and bring a printed copy to class.

1. Go to UC Santa Barbara's [presidential speeches data repository](#) and pick a State of the Union Address given by a president of your choice. Download the text file as a `.txt`. (you can copy paste the text contents into a notepad editor and save as a `.txt` file) Once you have the `.txt` file ready, read it into Python.
2. Make all words lowercase, trim stopwords and punctuations. Retrieve only the stems of the words such that words like family and families would be considered as same.
3. Create a barchart showing the top 10 words used in the Address.
4. Create two side-by-side barcharts showing the top 10 positive and negative words used in the Address.
5. Make a word cloud showing the most frequent words used in the Address.
6. Taken together, why do you think the president chose to use these words you see on your barcharts frequently? Are you surprised by any of the frequent words you identified? Why or why not?